

Christian Tavera

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Education

Massachusetts Institute of Technology: Candidate for B.S. in Mechanical Engineering, Minor in Design - GPA: 4.6 **May 2027**

- Relevant Courses: Design & Manufacturing II, Thermal-Fluids Engineering II, Mechanics & Materials II, Dynamics & Control II, Design Studios, Design Computation, Measurement & Instrumentation, Electronics for Mechanical Systems, Mechanical Engineering Tools, Tool Design

Skills

Computer-Aided Design: SOLIDWORKS, Fusion 360, PCB Design (KiCad), Circuit Design (LTspice), Rhino, Grasshopper, LightBurn

Programming: Python, MATLAB, Git, C++, React, HTML, CSS, JavaScript

Fabrication: DFMA, CNC Mill, Lathe, Injection Mold, Waterjet, Laser Cutter, FDM/SLA, Blast Cabinet, Thermoform, Sheet Metal Forming

Experience

University of Melbourne Department of Mechanical Engineering – Networked Systems Laboratory **Melbourne, VIC**

Robotics Researcher

June 2025 - August 2025

- Designed and implemented hybrid mechanical-controls platform, facilitating multi-agent tracking in high-precision applications
- Led design of a low-cost, quick-installation suspension system for a fleet of omnidirectional robots, expanding demos beyond padded terrain

MIT Computer Science & Artificial Intelligence Laboratory – Marine Robotics Group

Cambridge, MA

Undergraduate Researcher

June 2024 - August 2024

- Designed and fabricated hydrophone array systems—with weight and drag constraints of AUV towing—for multi-agent acoustic localization
- Pitched design revisions to research team, wrote grant proposals, and streamlined project timeline to reduce constraints of quotes and shipments
- Analyzed array apertures by creating visualizations—successfully doubling the bearing resolution of the array to enhance localization accuracy

MIT Motorsports

Cambridge, MA

Electrical-Mechanical Integration Engineer

September 2023 - May 2024

- Developed watertight boxes to house high-voltage circuit elements, and collaborated on designs for a user-centric steering-wheel layout
- Facilitated collaboration and established a point of contact between electrical and mechanical subteams to ensure effective systems engineering

Mineola Summer Robotics Intensive

Mineola, NY

Founder and Lead Instructor of Summer STEM Intensive

June 2023 - August 2023

- Created program and curriculum, mentoring students ages 9-15 on circuit design and software fundamentals with capstone Arduino project
- Fostered student engagement and enrollment, managed team of TAs, proposed summer program to school admin to obtain space and funding

Projects

Walkman-Inspired Yo-Yos (50+ Unit Run) | DFMA, Injection Molding, Overmolding, CNC Machining, SLA, CAM, Drop Testing

- Co-developed 10-part injection molded assembly with an overmold, thermoformed windows, insert-molded hardware, and cosmetic features
- Iterated CNC and SLA tooling for robust press-fit assembly, validated through 5 ft drop testing; characterized process for 50+ unit production run

1st Place Architected Nanomaterial Competition | FEA, SolidWorks, Two-Photon Polymerization, Compression Testing

- Submitted top-performing geometry across 125 entries from 39 teams, optimizing elastic modulus to density ratio via iterative FEA
- Fabricated unit cells with 5µm features using MIT.nano's two-photon polymerization, validating simulated stiffness and toughness

Compliant Skateboard Bushing Research, MIT Sports Summit 2026 | Product Characterization, Technical Writing

- Designed method to isolate bushing tightness; quantifying correlations of elastic stiffness and viscoelastic damping in turning system
- Authored paper & poster accepted to MIT Sports Summit, translating product performance data for mixed technical audiences

Distributed Controls Systems | SolidWorks, Arduino, C++, MATLAB, Manual Machining

- Implemented and tuned PID controls system with dead-zone and slip detection, validated with precise multi-agent formations

Hydrophone Array for Underwater Vehicle Localization | SolidWorks, KiCad, LTspice, Python, MATLAB, CNC Machining

- Modeled 3D-printed housing for hydrostatic pressures, designed and simulated high-frequency circuitry, and programmed SPI protocol
- Designed an array sensitivity simulation. Parameterized inputs for user-centric frontend, enabling researchers to quickly optimize array spacings
- Authored documentation of project insights and research in SLAM and acoustics applications to foster further research utilizing the array